



Feed the Future Innovation Lab for Livestock Systems

Campylobacter Genomics & Environmental Enteric Dysfunction (CAGED) Project & Exposure Assessment of CAMpylobacter (EXCAM)

Standard Operating Procedure (SOP) for Environmental Sample Collection

SOP # C3.3C

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Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21,2020
Page 2 of 18			

Table of Contents

<i>SOP Purpose</i>	3
<i>Risk Assessment and Safety Measures</i>	3
<i>Definitions, Abbreviations and Acronyms</i>	4
<i>Responsibilities</i>	4
<i>References</i>	5
<i>Procedure</i>	5
A. Record Equipment	6
B. Record Materials, Supplies, and Reagents	6
C. Preparation	7
D. Routine Disinfection in the field	9
E. Collection of the samples in the field	10
Soil Sampling	10
Drinking Water/Bathing Water Sampling	11
Food Sampling.....	12
Fomite Sampling	13
F. Length of Procedure	14
G. Sample Transportation	14
H. Receiving of the samples at HU	14
<i>Emergency Procedures</i>	14
<i>Records Management</i>	14
<i>Revision History/ Rationale</i>	15

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21, 2020
Page 3 of 18			

Experimental Location(s): Ethiopia

SOP Purpose

The purpose of this document is to describe the technical procedures required to collect representative environment samples from the selected cohorts during the "Campylobacter Genomics and Environmental Enteric Disfunction "(CAGED) longitudinal study and "Exposure Assessment of *Campylobacter* Infections in Rural Ethiopia" (EXCAM) cross sectional study. This SOP will outline the different methods of sample collection based on the type of sample being collected for sample processing, provide sample collectors with information on equipment, materials and data sheets needed for sample collection and personal protective equipment (PPE) requirements.

Risk Assessment and Safety Measures

Each SOP should include a detailed outline of the potential safety hazards to ensure the safety of all team members, participants, and samples by mitigating the listed potential safety hazards. An example of safety procedures should include using proper personal protective equipment (PPE) such as gloves, shoe covers, or a face mask.

RISK ASSESSMENT: *Identify potential safety hazards.*

EXPOSURE FACTORS	ENVIRONMENT FACTORS	ANIMAL/PLANT FACTORS	PROCESS RISKS
☐ Oral transmission route	☐ Environmental stability	☐ Amplifying host	☐ Cryogen
☐ Percutaneous transmission route	☐ Reproducible agent/vector	☐ Agent shedding	☐ Corrosives
☐ Mucosal transmission route	☐ Endemic host/vector	☐ Physical hazards (teeth, claws, etc.)	☐ Flammables
☐ Inhalation transmission route	☐ Drug-resistant pathogen	☐ Exotic organism	☐ Oxidizers
☐ Person-to-person communicable	☐ Human pathogen	☐ Noxious / invasive organism	☐ Carcinogens

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21, 2020
Page 4 of 18			

☐ Fomite	☒ Animal pathogen	☒ Field study	☐ Radionucleotides
	☐ Plant pathogen		

PERSONAL PROTECTIVE EQUIPMENT:

☐ Re-usable lab coat/gown	☐ Surgical mask
☒ Disposable lab coat/gown/coverall	☐ N95 respirator
☒ Disposable (latex/nitrile) gloves	☐ Powered Air Purifying Respirator (PAPR)
☒ Safety glasses/goggles	☐ Heat-resistant gloves
☐ Face shield	☐ Cold-resistant gloves
☐ Hearing protection	☒ Other: Bug spray and Sun glasses

ENGINEERING CONTROLS:

☐ Biological safety cabinet (BSC)	☐ Aerosolization chamber
☐ Fume hood	☐ Autoclave
☐ Cages/Isolators/Growth chambers	☐ Hand-washing sink
☐ Facility directional airflow	☐ Separation of rooms/ functions (i.e. for PCR work)
	☒ Other: Hand Sanitizer

Definitions, Abbreviations and Acronyms

No definitions, Abbreviations or Acronyms have been identified for the purpose of this SOP at this time.

Responsibilities

All designated team members are required to read the SOP in detail before executing the specified procedure. Once the team member has read the SOP at length, they will be required to sign the SOP verification sheet under records management (section 8) within the SOP. The project manager or team leader will verify everyone's knowledge and training of the SOP before procedure execution. It is the responsibility of both the staff, project manager and team leaders to ensure that the procedure or process follows the details noted in the individual SOP and to detail in writing when the SOP or a component of that SOP has not been followed along with reasoning for SOP deviation.

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21, 2020
Page 5 of 18			

Additional Responsibilities

- All sample collectors performing sample collection practices must adhere to the safety practices and PPE listed above.
- All sample collectors must fill out all sample collection logs at the site of collection.

References

Water sampling methods were developed by utilizing the Sanipath Environmental Sampling Protocols as a guideline. More additional information regarding water sampling can be found on the Sanipath website

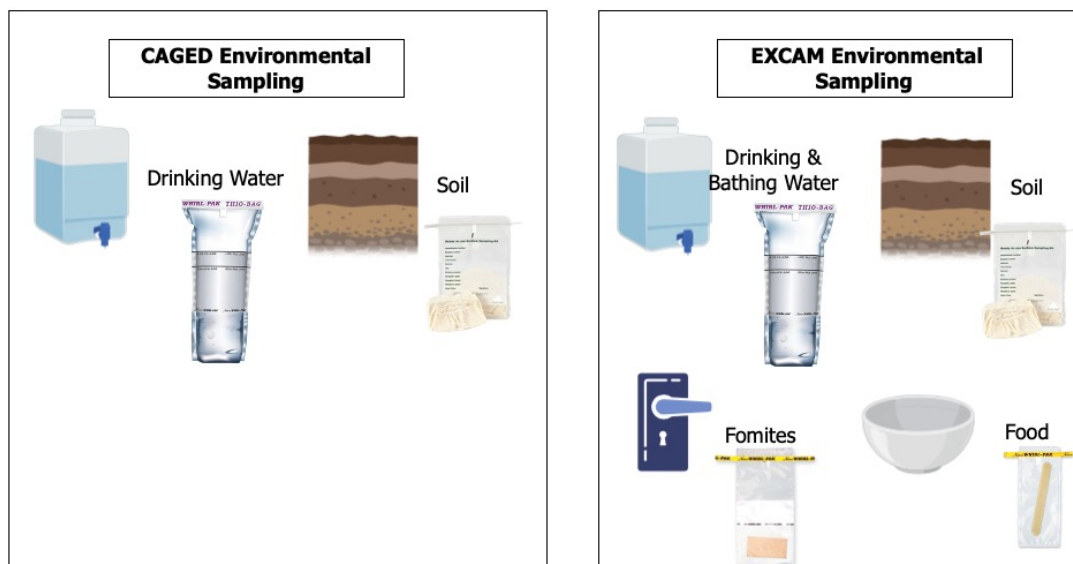
<https://sites.google.com/view/sanipathwiki/environmental-sample-collection/environmental-sample-collection-downloads?authuser=0>

Procedure

The CAGED and EXCAM project will both require sampling of various environmental samples. CAGED aims to enroll a total of 120 children living in selected household from six cohorts (20 participants per cohorts). Out of the 120 selected household, two types of environmental (soil and water) samples will be collected from each household once every six months. EXCAM will enroll a total of 78 participants based on the existing participants enrolled in CAGED (a total 13 participants per cohort [n=6]). EXCAM will involve the collection of the following environmental samples (drinking water, bathing water, soil, fomites, and food). Details on the sampling plan can be found in the weekly sampling schedule located on Microsoft Teams. A summary of the types of samples collected for each project can be found in **Figure 1**.

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21, 2020
Page 6 of 18			

Figure 1. CAGED & EXCAM Environmental Sampling Summary



A. Record Equipment

Item	Location	Special Notes (catalog #, vendor)
Balance	Main Lab Room	Used for weighing animal fecal samples
Cooler boxes, Ice packs, and Water	Sample Receiving Room	Used for sample transportation
Foldable table	Sample Receiving Room	Workspace for sample collection (if needed)

B. Record Materials, Supplies, and Reagents

Item	Information/Location	Purpose (details can be found in section E)
Boot Socks	VWR/Solar Biological Cat# SH-BTSW	Soil Sample Collection
Skim Milk Powder	VWR Cat# IC90288705	Preparation: To enrich boot socks for soil sample collection

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21,2020
Page 7 of 18			

1-Liter container	VWR Cat # 89166-108	Drinking Water Sample
Whirl-pak bags	VWR Cat# 66175-612	Food Sample
Sterile Tongue Depressors		Assist with food sample collection
Drag swabs/swabs		Fomite Sample
Permanent Marker	VWR Cat# 500034-974	Labeling
Tube racks		To hold sample tubes during collection and field transport
Gloves		PPE
Hand Sanitizer		Hand washing
Bug spray/insecticide		Protect sample collector from mosquitoes
Trash bag		Workspace set up
70% Ethanol		Disinfecting work surface area and tools
5% sodium hypochlorite		Disinfecting work surface area and tools
Paper Towels		Disinfecting work surface area and tools

C. Preparation

Sample collection (n=5-7 for EXCAM; soil, drinking water and bathing water, food, and fomite (x1-3 depending on the observations made by the field team); n=4 for CAGED; three soil and one water sample) will require the use of a variety of materials that will be prepared in advance by lab personnel (**Table 1**). Sample collectors will collect prepared materials for sample collection either the day of sample collection or the day before and verify with **Appendix 1. Sample Collection Materials Checklist** before starting field work.

Table 1. Collection of the environmental samples

Sample type	Preparation	Storage condition during transport
Soil	Wetted (skim milk) boot socks (4 per household)	Ice cooler containing water and ice packs

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21, 2020
Page 8 of 18			

Water	1-Liter autoclavable water bottle	Ice cooler containing water and ice packs
Food	Whirl-pak bags and sterile tongue depressors	Ice cooler containing water and ice packs
Fomite	Drag swabs, sterile PW, and 50 ml tubes	Ice cooler containing water and ice packs

Laboratory personnel will prepare the sample collection materials in advance to ensure that the collection tubes and bags have proper reagents for sample transportation. Materials will be subdivided into designated project bins to assist with organization during field work. All materials will be labeled by lab personnel with the name of reagent/medium and color coded (Blue for CAGED and Green for EXCAM) and pre-labeled with printed sample labels. A visual representation of the sample labeling system can be found in **Figure 2** and details on the sample labeling scheme can be found in **Table 2**. **It is important that the laboratory personnel prepared extra supplies (approx. 25%) for sample collection to assure that the sample collectors have enough supplies in case they have to redo the sampling procedure for some of the samples.**

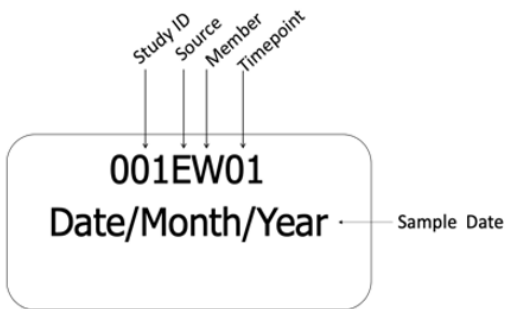
Figure 2. Environmental Sample Label		Table 1. Environmental Member ID Summary																																	
		<table> <tr> <th>Sample Type</th><th>Member ID</th><th>Sample Description</th><th>Project</th></tr> <tr> <td rowspan="3">Water</td><td>W</td><td>Drinking Water</td><td>CAGED</td></tr> <tr> <td>D</td><td>Drinking Water</td><td>EXCAM</td></tr> <tr> <td>B</td><td>Bathing Water</td><td>EXCAM</td></tr> <tr> <td rowspan="3">Soil</td><td>S(1)</td><td>Soil (Inside Household 1)</td><td>CAGED</td></tr> <tr> <td>S(2)</td><td>Soil (Inside Household 2)</td><td>CAGED</td></tr> <tr> <td>S(0)</td><td>Soil (Front House)</td><td>CAGED</td></tr> <tr> <td>Fomite</td><td>V</td><td>Fomite</td><td>EXCAM</td></tr> <tr> <td>Food</td><td>F</td><td>Food</td><td>EXCAM</td></tr> </table>		Sample Type	Member ID	Sample Description	Project	Water	W	Drinking Water	CAGED	D	Drinking Water	EXCAM	B	Bathing Water	EXCAM	Soil	S(1)	Soil (Inside Household 1)	CAGED	S(2)	Soil (Inside Household 2)	CAGED	S(0)	Soil (Front House)	CAGED	Fomite	V	Fomite	EXCAM	Food	F	Food	EXCAM
Sample Type	Member ID	Sample Description	Project																																
Water	W	Drinking Water	CAGED																																
	D	Drinking Water	EXCAM																																
	B	Bathing Water	EXCAM																																
Soil	S(1)	Soil (Inside Household 1)	CAGED																																
	S(2)	Soil (Inside Household 2)	CAGED																																
	S(0)	Soil (Front House)	CAGED																																
Fomite	V	Fomite	EXCAM																																
Food	F	Food	EXCAM																																

Figure 2. Represents an example of the environmental sample labeling scheme that will be utilized by CAGED & EXCAM. The Study ID will be based on participant enrollment and the Source ID (E) represents it is an environmental sample.

All food sample bags must be labeled with the sample ID, type of sample, collection date, and project. Verify that the label matches the sample type and ID as indicated

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21,2020
Page 9 of 18			

by the project manager prior to sampling. This information must be recorded, to track collected samples. Please review **Appendix 2** for more details on **Environmental Sample Collection Log**.

D. Routine Disinfection in the field

The field team must wear appropriate PPE (gloves, lab jacket, and shoe covers; listed in section 2) and set up the workspace before collecting samples in the designated household. Disinfection of the field team (hand and boots) and tools must be performed before and after collecting a sample in the designated household. Similarly, the field team and tools must be disinfected once arriving and before leaving a household.

The disinfection of the field team and tools must be performed using 70% ethanol and paper towels. Allow the disinfected area/tools to dry before starting the sampling procedure.

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21,2020
Page 10 of 18			

E. Collection of the samples in the field

Below you will find the detailed methods for sample collection associated with each specific type of sample.

Soil Sampling (step by step)

Soil samples will be collected for the purpose of CAGED (3x) and EXCAM (1x).

The following steps must be followed to collect the CAGED soil samples:

1. Using clean sterile gloves, put on the clean shoes dedicated for the soil sampling (this step must be performed where the soil sampling will be performed)
2. Using clean sterile gloves, put on the shoe covers (which is already pre-moistened with double-strength skim milk) over clean shoes (this step must be performed where the soil sampling will be performed)
3. Walk around the property (preferably a 3 x 5 m area; where children have access; 15 steps; making sure to cover most of the area; Figure 3). **The collectors will walk like a "blind man".**
4. Using clean sterile clothes, place both shoes covers back into the corresponding labeled sample bag.
5. Remove the air from the bag and seal it by folding and twisting the ends of the bags.
6. Store the bags in a cooler box containing water and ice packs. Once at the laboratory, **store them at 4°C.**
7. Record samples in **Appendix 2 Environmental Sample Collection Log.**

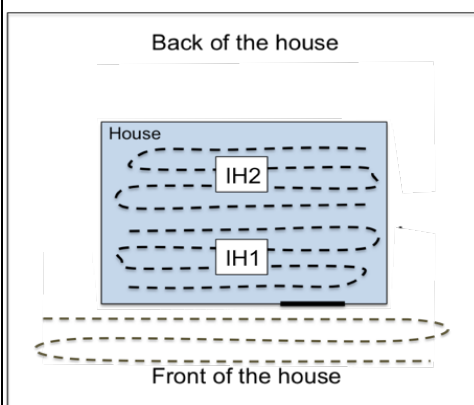


Figure 3. Collection of environmental samples. Three samples will be collected per house: two samples inside the house (IH1 and IH2), one sample in front of the house (FH). Dotted black line: the 15 steps that the collector will perform to collect the samples.

Disinfection of the shoes used for soil sampling: Thoroughly clean shoes with a brush, soap, and sterile water between houses and at the end of the day. Disinfect soles with alcohol. Put

shoes on while standing on a clean sheet brought from the lab. Never walk with these shoes without the shoe covers.

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21,2020
Page 11 of 18			

Drinking Water/Bathing Water Sampling (step by step)

Two types of water samples will be collected. Drinking Water (1x for CAGED and 1x for EXCAM) and bathing water (1x for EXCAM) will be collected.

The following steps must be followed to collect the drinking water:

1. Gown on PPE (listed in section 3). Put on gloves and spray your hands with 70% ethanol. Allow gloves to dry before proceeding.
2. Verify that the Whirl-pak bags are correctly labeled.
3. If the main source of drinking water is from a pipe, carefully unscrew the Whirl-pak bag and place it a few inches away from the tap, pump, or dispenser.
4. If the main source of drinking water is from a borehole, storage container, or dug well. Ask the participant to assist with sample collection by collecting water as they typically do. Once the water has been collected, carefully fill the Whirl-pak bag.
5. Close the Whirl-pak bags. Confirm that all labels are still in place.
6. Collect at least 1 liter of water per household
7. Store the Whirl-pak bags in a cooler box containing ice packs and water during the field sampling. Once at the laboratory, **store them at 4°C.**
8. Record samples in **Appendix 2 Environmental Sample Collection Log.**

Once the bathing water has been identified, similar sampling procedures used for the drinking water must be used for the bathing water.

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21,2020
Page 12 of 18			

Food Sampling (step by step)

Food samples will be collected for the purpose of EXCAM only (x1-2). Food samples are designated as complementary food (porridge, biscuit, ...) fed to the children other than breast milk.

1. The mother/caretaker, with the help of the sample collector, will transfer at least 10 grams of food into a Whirl-pak bag with the corresponding sample ID using the spoon or other kitchen tool used to feed the child.
2. Remove the air from the bag and seal it by folding and twisting the ends of the bags.
3. Store the container in a cooler box containing water and ice packs.
4. Record samples in **Appendix 2 Environmental Sample Collection Log**. Please indicate what type of food sample was collected in the "remarks section" of the **Appendix 2 Environmental Sample Collection Log**.

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21,2020
Page 13 of 18			

Fomite Sampling (step by step)

Fomite samples (between 1-3x per household) will be collected for the purpose of EXCAM only. Fomite sampling will be dependent on infant behavioral observation. Thus, there must be clear communication between the observation and field team.

1. Carefully remove the pre-wetted sterile swab from the container and place the swab on the designated surface area or object. Move the swab back and forth at least five times.
2. Turn the swab and repeat to ensure the entire swab has made contact with the surface area or object.
3. Carefully place the swab into its original container.
4. Remove the air from the bag and seal it by folding and twisting the ends of the bags.
5. Store the container in a cooler box containing water and ice packs
6. Record samples in **Appendix 2 Environmental Sample Collection Log**. Please indicate what type of object/area was sampled in the "remarks section" of the **Appendix 2 Environmental Sample Collection Log**.

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21, 2020
Page 14 of 18			

F. Length of Procedure

The length of the procedure for sample collection will be impacted by the type of sample. Thus, the length of procedure is an estimation of the process rather than an exact time.

Type of Sample	Method	Time
Environmental (Soil)	Boot sock	10-15 minutes
Environmental (Drinking Water/Bathing Water)	1-Liter Container	10-15 minutes
Environmental (Food)	Whirl-pak bags	5-10 minutes
Environmental (Fomites)	Drag Swabs	5-10 minutes

G. Sample Transportation

All samples collected in the field will be stored in a cooler box that contains ice packs and water during field work and transportation to the lab for processing.

H. Receiving of the samples at HU

All samples will be received in the lab by lab personnel and recorded using the sample logbook. Field collection recordings will be scanned and saved electronically.

Emergency Procedures

In case of emergencies during field collection please contact the project manager immediately at phone number +251 91211 5740. All emergencies must be documented and reported to the principal and co-investigators in a timely manner.

Records Management

A. SOP LOCATION:

A hard copy of the SOP will be kept in the sample collection log binder to be easily accessed in the field. An additional hard copy will be kept in the lab for reference.

B. TRAINING/ SOP VERIFICATION SHEET:

Each team member shall receive training at the time of initial assignment to the lab and prior to assignments involving new procedures. This training shall include, but not be limited to the hazards identified in the Risk Assessment, extensive review of the SOP, measures team members must follow to protect themselves from exposure to these hazards, and availability of reference

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21,2020
Page 15 of 18			

	materials, such as Material Safety Data Sheets. All training and review of SOP by the team member will be recorded in the SOP Distribution List The SOP Distribution List must be completed and saved as a record of the training and review of SOP. If the Project Manager/ Team Leader/ or PI/ CoPI feel that the team member requires an SOP refresher training the team member will need to complete the SOP Verification form again to document the refresher training.
	Revision History/ Rationale No revision present at this time.

Environmental Sample Collection			
SOP No. C3.3C	Version 1.0	Status Effective	Effective Date December 21,2020
Page 16 of 18			

Appendix 1. Sample Collection Materials Checklist

Sampling Date: _____

Sampling Locations: _____

Additional Remarks: _____

Sample Collection Materials Checklist				
Materials	Quantity	Prepared by (Lab Initials)	S. Collector Initials	Remarks
Boot socks (Soil)				
1-Liter sterile bottles (Water)				
Whirl-pak bags (Food)				
Sterile tongue depressors				
Drag swabs				
Bottles of 1XPBS or PW				
Cup				
Tube racks				
Gloves				
Methylated spirits or other disinfectant				
Permanent marker				
Metals spatulas				
Paper Towels				

Appendix 2. Environmental Sample Collection Log

Sampling Date: _____ Sample Collector(s): _____ Sampling Location: _____

Lab Sample Arrival Time: _____ Person receiving the samples in the lab: _____

Environmental Sample Collection Tracking Log											
No.	Household ID	Environmental Sample ID								Sampling Time	Remarks
		Soil	Soil Type	Water	Water Type	Food	Food Type	Fomite	Fomite Type		
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											

Please note that all sample collectors must record the following information:

No. represent the number of sample vials used.

Date represents the date of sample collection.

Household ID are assigned by the project manager.

Environmental Sample Collection ID represents the type of environmental sample that was collected. Please only enter in sample information in one row for the Environmental Sample Collection ID section. For soil, please indicate if this was inside household 1 (IH1), inside household 2 (IH2) or soil from the front of the house (FH). For water, please indicate whether drinking or bathing water was collected and indicate the source of water supply in the remarks section. For fomite sampling please indicate which fomite was collected. Finally, for food, please indicate which type of food sample was collected.

Sampling time represents the time in which the samples were collected.

Remarks represents any additional comments that is important to mention that occurred during sampling.